Mark Hanna

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Education

Western University London, ON

Bachelor of Engineering Sciences (BESc), Software Engineering *Expected Graduation: May 2029*

- **Relevant Coursework:** Database Management Systems, Requirements Analysis and Engineering

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C, C++, C#, HTML, CSS, VHDL, MATLAB
Web & Back End: React, Node.js, Express, REST APIs, PostgreSQL, Supabase, Tailwind CSS, Vite
DevOps & Tools: Git, GitHub Actions, CI/CD, Docker, Vercel, Postman, Vitest, VS Code, Unity
Embedded & Hardware: STM32, STM32CubeIDE, FPGA, Quartus Prime, Intel MAX10, CAN, UART, Photonic ICs
Design & Simulation: SolidWorks, Onshape, Simulink, Oscilloscopes, Excel VBA

Experience

Back End Developer May 2025 - Sep. 2025

RM Health Supplies *Mississauga, ON*

- Engineered **REST APIs** with **Node.js, Express,** and **SQL** that supported **5 internal tools**.
- Implemented indexing and query restructuring in **PostgreSQL**, raised throughput by **25%**.
- Diagnosed a **memory leak** in a long-running service, cut crash frequency from **4 per week to 0**.
- Deployed background jobs with **cron** and **queue workers**, cut manual admin tasks by **60%**.
- Added **input validation** and **logging middleware**, reduced error rates by **15%**.

Computer Technician Sep. 2023 - Dec. 2024

Technoid Gaming Computers *Mississauga, ON*

- Produced assembly instructions and diagrams, reduced build time by **20%**.

Projects

World Cup 2026 Predictor May 2026 - Jul. 2026

Team Project (2 members) *React, Supabase, PostgreSQL, Python, GitHub Actions*

- Co-built a full-stack **React** and **Supabase** app used by **120+ players** in **8 leagues**, with auth and live brackets.
- Trained a **logistic regression** model on **2,448 matches**, reached **68.6%** cross-validated accuracy.
- Automated daily model refreshes and **10,000**-run **Monte Carlo** simulations with **GitHub Actions**.
- Deployed a **Supabase Edge Function** on **pg_cron** for minute-by-minute live scores over **Realtime**.
- Set up **CI/CD** with lint, **Vitest** unit tests, and build checks on every pull request.

Dobby's Night Out Apr. 2026

Team Project (5 members) *C#, Unity 6, Git, Object-Oriented Design*

- Prototyped the Cat Smack ability in **C#** for a **5-level Unity 6** RPG built by **5 students** in **3 weeks**.

4-Bit ALU on FPGA Jun. 2025

Personal Project *VHDL, FPGA, Hardware Verification, RTL Design*

- Engineered a 4-bit ALU with **28 logic gates** and a **4-bit ripple-carry adder**, validated at **50 MHz**.

Fall Detection System with STM32 Nov. 2024

Personal Project *C/C++, Microcontrollers, SolidWorks, STM32CubeIDE*

- Implemented accelerometer sampling at **200 Hz** with **C++** on **STM32**, using threshold-based logic.
- Delivered GSM alerts in under **3 s** with a **92%** true-positive detection rate.

Extracurricular

Western Baja Racing Sep. 2025 - Present

Western University *London, ON*

- Implemented **UART-based data acquisition** on **STM32**, processed **500+ samples per second**.
- Optimized embedded filtering logic, lowered sensor read errors by **30%** during vibration tests.
- Helped deploy a **Python dashboard** with real-time plots, cut analysis time by **40%**.